

US Patent & Trademark Office

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United States Patent Application Publication

20250286208

Kind Code

A1

Publication Date

September 11, 2025

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SECONDARY BATTERY AND SECONDARY BATTERY MODULE INCLUDING THE SAME

Abstract

A secondary battery configured to prevent damage around a cell vent and prevent heat propagation and thermal runaway and a secondary battery module including the same. The secondary battery includes a case having a space, an electrode assembly accommodated in the space of the case, a cap assembly including a cap plate configured to seal the space of the case, the cap plate including a vent unit in one surface of the cap plate, and a terminal coupled to one side of the cap plate, the terminal being electrically connected to the electrode assembly, and an insulating member coupled to the surface of the cap plate, the insulating member being configured to cover at least a part of the vent unit.

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Family ID: 96949906

Appl. No.: 18/778295

Filed: July 19, 2024

Foreign Application Priority Data

KR

10-2024-0033611

Mar. 11, 2024

Publication Classification

Int. Cl.: H01M50/358 (20210101); H01M50/103 (20210101); H01M50/15 (20210101)

U.S. Cl.:

CPC H01M50/358 (20210101); H01M50/103 (20210101); H01M50/15 (20210101);

Background/Summary

CROSS-REFERENCE TO THE RELATED APPLICATION

[0001] The present application claims priority to and the benefit of Korean Patent Application No. 10-2024-0033611, filed on Mar. 11, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

1. Field

[0002] Embodiments relate to a secondary battery and a secondary battery module including the same.

2. Description of the Related Art

[0003] Unlike primary batteries that are not designed to be recharged, secondary (or rechargeable) batteries are batteries that are designed to be discharged and recharged. Low-capacity secondary batteries are used in portable, small electronic devices, such as smart phones, feature phones, notebook computers, digital cameras, and camcorders, while large-capacity secondary batteries are widely used as power sources for driving motors in hybrid vehicles and electric vehicles and for storing power (e.g., home and/or utility scale power storage). A secondary battery generally includes an electrode assembly including a positive electrode and a negative electrode, a case accommodating the positive and negative electrodes, and electrode terminals connected to the electrode assembly.

[0004] The above information disclosed in this Background section is for enhancement of understanding of the background of the present disclosure, and therefore, it may contain information that does not constitute related (or prior) art.

SUMMARY

[0005] Embodiments provide a secondary battery configured to prevent (or at least mitigate) damage around a cell vent and to prevent (or at least mitigate) heat propagation and thermal runaway and a secondary battery module including the secondary battery.

[0006] These and other aspects and features of the present disclosure will be described in or will be apparent from the following description of embodiments of the present disclosure.

[0007] A secondary battery according to an embodiment of the present disclosure to accomplish the above object includes a case having a space defined therein, an electrode assembly accommodated in the space of the case, a cap assembly including a cap plate configured to seal the space of the case, the cap plate including a vent unit in one surface thereof, and a terminal coupled to one side of the cap plate, the terminal being electrically connected to the electrode assembly, and an insulating member coupled to the surface of the cap plate, the insulating member being configured to cover at least a part of the vent unit.

[0008] In some examples, the vent unit may be coupled to a lower surface of the cap plate, and the insulating member may be coupled to an upper surface of the cap plate and correspond to the position of the vent unit.

[0009] In some examples, the vent unit may be coupled to the cap plate and seal a vent hole formed in one side of the cap plate, and the vent unit may include a connection portion overlapping the cap plate, the connection portion being fixed to the cap plate, and a rupture portion inside the

connection portion, the rupture portion having a notch.

[0010] In some examples, the insulating member may have an opening in the center region of the insulating member to expose the vent unit.

[0011] In some examples, the insulating member may include an insulating plate coupled to the upper surface of the cap plate and a protrusion protruding from the insulating plate toward the vent unit.

[0012] In some examples, a shape of the protrusion may correspond to a shape of the vent hole.

[0013] In some examples, the protrusion may have a hollow, columnar structure corresponding to the shape of the vent hole.

[0014] In some examples, the opening may extend through the insulating plate and the protrusion.

[0015] In some examples, the insulating member may be coupled along the edge of the vent hole through the protrusion.

[0016] In some examples, the cap plate may further include a seating recess in the upper surface of the cap plate around the vent hole.

[0017] In some examples, the insulating plate may be coupled to the seating recess.

[0018] In some examples, the insulating member may be coupled to the seating recess of the cap plate by an interference fit.

[0019] In some examples, the insulating member may have a ring-shaped top structure with a hollow interior.

[0020] In some examples, the insulating member may be coupled to the upper surface of the cap plate by thermal fusion.

[0021] In some examples, the insulating member and the cap plate may be integrally formed by double injection molding.

[0022] In some examples, an upper surface of the insulating member and the upper surface of the cap plate may be substantially co-planar.

[0023] In some examples, the insulating member may include at least one of polyphenylene sulfide (PPS), polyetheretherketone (PEEK), or mica.

[0024] In some examples, in response to the rupture portion being opened by the internal pressure of the case, the rupture portion may rupture along the edge of the opening of the insulating member.

[0025] In some examples, the insulating member may be configured to cover the entirety of the vent unit.

[0026] A secondary battery module according to an embodiment of the present disclosure to accomplish the above object includes secondary batteries arranged in one direction, a housing having sidewalls configured to form a receiving portion configured to accommodate the secondary batteries, and a busbar configured to electrically connect neighboring ones of the secondary batteries to each other, wherein each of the secondary batteries includes a case having a space defined therein, an electrode assembly accommodated in the space of the case, a cap assembly including a cap plate configured to seal the space of the case, the cap plate including a vent unit in one surface thereof, and a terminal coupled to one side of the cap plate, the terminal being electrically connected to the electrode assembly, and an insulating member coupled to the surface of the cap plate, the insulating member being configured to cover at least a part of the vent unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The following drawings attached to this specification illustrate embodiments of the present disclosure, and further describe aspects and features of the present disclosure together with the detailed description of the present disclosure. Thus, the present disclosure should not be construed

as being limited to the drawings:

[0028] FIG. 1A is a perspective view of a prismatic battery according to an embodiment of the present disclosure;

[0029] FIG. 1B is a cross-sectional view taken along the line **1b-1b** in FIG. 1A.

[0030] FIG. 2 is a perspective view of a battery module according to an embodiment of the present disclosure;

[0031] FIGS. 3A and 3B are perspective views of a battery pack according to an embodiment of the present disclosure;

[0032] FIGS. 4A and 4B show examples of vehicle bodies and body parts to which one or more embodiments of the present disclosure may be applied;

[0033] FIGS. 5A to 5D illustrate a method for charging a secondary battery according to an embodiment of the present disclosure;

[0034] FIG. 6 is a partial exploded perspective view of a vent unit and an insulating member of a secondary battery according to an embodiment of the present disclosure;

[0035] FIG. 7A is a partial sectional view of the vent unit and the insulating member of the secondary battery according to the embodiment of the present disclosure;

[0036] FIG. 7B is a sectional view showing the vent unit of FIG. 7A when ruptured;

[0037] FIG. 8 is a partial exploded perspective view of a vent unit and an insulating member of a secondary battery according to another embodiment of the present disclosure; and

[0038] FIG. 9 is a partial sectional view of the vent unit and the insulating member of the secondary battery according to the other embodiment of the present disclosure.

DETAILED DESCRIPTION

[0039] Hereinafter, embodiments of the present disclosure will be described, in detail, with reference to the accompanying drawings. The terms or words used in this specification and claims should not be construed as being limited to the usual or dictionary meaning and should be interpreted as meaning and concept consistent with the technical idea of the present disclosure based on the principle that the inventor can be his/her own lexicographer to appropriately define the concept of the term to explain his/her invention in the best way.

[0040] The embodiments described in this specification and the configurations shown in the drawings are only some of the embodiments of the present disclosure and do not represent all of the technical ideas, aspects, and features of the present disclosure. Accordingly, it should be understood that there may be various equivalents and modifications that can replace or modify the embodiments described herein at the time of filing this application.

[0041] It will be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected, or coupled to the other element or layer or one or more intervening elements or layers may also be present. When an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. For example, when a first element is described as being “coupled” or “connected” to a second element, the first element may be directly coupled or connected to the second element or the first element may be indirectly coupled or connected to the second element via one or more intervening elements.

[0042] In the figures, dimensions of the various elements, layers, etc. may be exaggerated for clarity of illustration. The same reference numerals designate the same elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Further, the use of “may” when describing embodiments of the present disclosure relates to “one or more embodiments of the present disclosure.” Expressions, such as “at least one of” and “any one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. When phrases such as “at least one of A, B and C,” “at least one of A, B or C,” “at least one selected from a group of A, B and C,” or “at least one selected from among A, B and C” are used to designate a list of elements A, B and C, the phrase may refer to any and all

suitable combinations or a subset of A, B and C, such as A, B, C, A and B, A and C, B and C, or A and B and C. As used herein, the terms “use,” “using,” and “used” may be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. As used herein, the terms “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art.

[0043] It will be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of example embodiments.

[0044] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” or “over” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein should be interpreted accordingly.

[0045] The terminology used herein is for the purpose of describing embodiments of the present disclosure and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0046] Also, any numerical range disclosed and/or recited herein is intended to include all sub-ranges of the same numerical precision subsumed within the recited range. For example, a range of “1.0 to 10.0” is intended to include all subranges between (and including) the recited minimum value of 1.0 and the recited maximum value of 10.0, that is, having a minimum value equal to or greater than 1.0 and a maximum value equal to or less than 10.0, such as, for example, 2.4 to 7.6. Any maximum numerical limitation recited herein is intended to include all lower numerical limitations subsumed therein, and any minimum numerical limitation recited in this specification is intended to include all higher numerical limitations subsumed therein. Accordingly, Applicant reserves the right to amend this specification, including the claims, to expressly recite any sub-range subsumed within the ranges expressly recited herein. All such ranges are intended to be inherently described in this specification such that amending to expressly recite any such subranges would comply with the requirements of 35 U.S.C. § 112(a) and 35 U.S.C. § 132(a).

[0047] References to two compared elements, features, etc. as being “the same” may mean that they are “substantially the same”. Thus, the phrase “substantially the same” may include a case having a deviation that is considered low in the art, for example, a deviation of 5% or less. In addition, when a certain parameter is referred to as being uniform in a given region, it may mean that it is uniform in terms of an average.

[0048] Throughout the specification, unless otherwise stated, each element may be singular or plural.

[0049] When an arbitrary element is referred to as being disposed (or located or positioned) on the “above (or below)” or “on (or under)” a component, it may mean that the arbitrary element is placed in contact with the upper (or lower) surface of the component and may also mean that another component may be interposed between the component and any arbitrary element disposed (or located or positioned) on (or under) the component.

[0050] In addition, it will be understood that when an element is referred to as being “coupled,” “linked” or “connected” to another element, the elements may be directly “coupled,” “linked” or “connected” to each other, or an intervening element may be present therebetween, through which the element may be “coupled,” “linked” or “connected” to another element. In addition, when a part is referred to as being “electrically coupled” to another part, the part can be directly connected to another part or an intervening part may be present therebetween such that the part and another part are indirectly connected to each other.

[0051] Throughout the specification, when “A and/or B” is stated, it means A, B or A and B, unless otherwise stated. That is, “and/or” includes any or all combinations of a plurality of items enumerated. When “C to D” is stated, it means C or more and D or less, unless otherwise specified.

[0052] A battery pack according to one or more embodiments includes at least one battery module and a pack housing having an accommodation space in which the at least one battery module is accommodated.

[0053] The battery module may include a plurality of battery cells and a module housing. The battery cells may be accommodated inside the module housing in a stacked form (or stacked arrangement or configuration). Each battery cell may have a positive electrode terminal and a negative electrode terminal and may be a circular type, a prismatic type, or a pouch type battery cell depending on the shape of battery. In the present specification, a battery cell may also be referred to as a secondary battery, a battery, or a cell.

[0054] In the battery pack, one cell stack may constitute one module stacked in place of the battery module. The cell stack may be accommodated in an accommodation space of the pack housing or may be accommodated in an accommodation space partitioned by a frame, a partition wall, etc. The battery cell may generate a large amount of heat during

[0055] charging/discharging. The generated heat may be accumulated in the battery cell, thereby accelerating the deterioration of the battery cell. Accordingly, the battery pack may further include a cooling member configured to remove the generated heat and thereby suppress deterioration of the battery cell. The cooling member may be at the bottom of the accommodation space where the battery cell is provided, but the present disclosure is not limited thereto and may be provided at the top or side depending on the battery pack.

[0056] The battery cell may be configured such that exhaust gas generated inside the battery cell under abnormal operating conditions, also known as thermal runaway or thermal events, is discharged to the outside of the battery cell. The battery pack or the battery module may include an exhaust port configured to discharge the exhaust gas and to thereby prevent or reduce damage to the battery pack or module by the exhaust gas.

[0057] The battery pack may include a battery and a battery management system (BMS) configured to manage the battery. The battery management system may include a detection device, a balancing device, and a control device. The battery module may include a plurality of cells connected to each other in series and/or parallel. The battery modules may be connected to each other in series and/or in parallel.

[0058] The detection device may be configured to detect a state of a battery (e.g., voltage, current, temperature, etc.) and to output state information indicating the state of the battery. The detection device may detect the voltage of each cell constituting the battery or of each battery module. The detection device may be configured to detect current flowing through each battery module constituting the battery module or the battery pack. The detection device may also be configured to detect the temperature of a cell and/or module on at least one point of the battery and/or an ambient

temperature.

[0059] The balancing device may be configured to perform a balancing operation of a battery module and/or cells constituting the battery module. The control device may be configured to receive state information (e.g., voltage, current, temperature, etc.) of the battery module from the detection device. The control device may be configured to monitor and calculate the state of the battery module (e.g., voltage, current, temperature, state of charge (SOC), life span (state of health (SOH)), etc.) based on the state information received from the detection device. In addition, based on the monitored state information, the control device may be configured to perform a control function (e.g., temperature control, balancing control, charge/discharge control, etc.) and a protection function (e.g., over-discharge, over-charge, over-current protection, short circuit, fire extinguishing function, etc.). In addition, the control device may be configured to communicate (in a wired or wireless manner) with an external device of the battery pack (e.g., a higher-level controller or vehicle, charger, power conversion system, etc.).

[0060] The control device may be configured to control a charging/discharging operation and a protection operation of the battery. To this end, the control device may include a charge/discharge control unit, a balancing control unit, and/or a protection unit.

[0061] The battery management system is a system that monitors the battery state and performs diagnosis and control, communication, and protection functions, and may calculate the charge/discharge state, calculate battery life or state of health (SOH), cut off, as necessary, battery power (e.g., relay control), control thermal management (e.g., cooling, heating, etc.), perform a high-voltage interlock function, and/or may detect and/or calculate insulation and short circuit conditions.

[0062] A relay may be a mechanical contactor that is turned on and off by the magnetic force of a coil or a semiconductor switch, such as a metal oxide semiconductor field effect transistor (MOSFET).

[0063] The relay control is configured to cut off the power supply from the battery in response to a problem occurring in the vehicle and the battery system and may include one or more relays and pre-charge relays at the positive terminal and the negative terminal, respectively.

[0064] In the pre-charge control, there is a risk of inrush current occurring in the high-voltage capacitor on the input side of the inverter when the battery load is connected. Thus, to prevent inrush current when starting a vehicle, the pre-charge relay may be operated before connecting the main relay and the pre-charge resistor may be connected.

[0065] The high-voltage interlock is a circuit that uses a small signal to detect whether or not all high-voltage parts of the entire vehicle system are connected and may have a function of forcibly opening a relay in response to an opening occurring at even one location on the entire loop.

[0066] FIGS. 1A is a perspective view illustrating a secondary battery according to one or more embodiments of the present disclosure, and FIG. 1B is a cross-sectional view taken along the line 1b-1b in FIG. 1A.

[0067] Referring to FIGS. 1A and 1B, the secondary battery **100** according to one or more embodiments of the present disclosure may include at least one electrode assembly **110** wound with a separator **113** as an insulator between the positive electrode **111** and the negative electrode **112**, a case **120** in which the electrode assembly **110** is received (or accommodated) therein, and a cap assembly **130** coupled to an opening of the case **120**.

[0068] The secondary battery **100** according to one or more embodiments illustrated in FIGS. 1A and 1B will now be described as a prismatic lithium ion secondary battery. However, the present disclosure is not limited thereto, and suitable aspects, features and principles described herein may be applied to various other types of batteries, such as lithium polymer batteries and/or cylindrical batteries.

[0069] Each of the positive electrode **111** and the negative electrode **112** may include a current collector made of a thin metal foil having a coated portion on which an active material is coated

and an uncoated portion **111a**, **112a** on which an active material is not coated (provided).

[0070] The positive electrode **111** and the negative electrode **112** are wound after positioning the separator **113**, which is an insulator, between the positive electrode **111** and the negative electrode **112**. However, the present disclosure is not limited thereto, and the electrode assembly **110** may have a structure in which a positive electrode **111** and a negative electrode **112**, each made of a plurality of sheets, are alternately stacked with a separator between the positive electrode **111** and the negative electrode **112**.

[0071] The case **120** may form or define the overall outer appearance of the secondary battery **100** and may be made of a conductive metal, such as aluminum, aluminum alloy, or nickel-plated steel. In addition, the case **120** may provide a space in which the electrode assembly **110** is accommodated.

[0072] The cap assembly **130** may include a cap plate **131** covering an opening in the case **120**, and the case **120** and the cap plate **131** may be made of a conductive material. The positive and negative electrode terminals **121** and **122** electrically connected to the positive electrode **111** and the negative electrode **112**, respectively, may penetrate (or extend through) the cap plate **131** and protrude outwardly therethrough.

[0073] In one or more embodiments, the positive and negative electrode terminals **121** and **122** protruding outwardly from the cap plate **131** may be coupled to the cap plate **131** by riveting or may be coupled to the cap plate **131** by welding. However, the present disclosure is not limited thereto, and an outer circumferential surface of an upper post of each of the positive and negative electrode terminals **121** and **122** protruding outwardly from the cap plate **131** may be threaded and may be fixed (coupled) to the cap plate **131** using a nut.

[0074] In addition, the cap plate **131** may be a thin plate and may be coupled to the opening in the case **120**, and an electrolyte injection port **132** into which a sealing stopper **133** may be installed may be located (e.g., formed) in the cap plate **131**, and a vent portion **134** having a notch **134a** may be installed.

[0075] The positive and negative electrode terminals **121** and **122** may be electrically connected to current collectors including first and second current collectors **140** and **150** (hereinafter referred to as positive and negative current collectors) by being bonded or coupled (e.g., by welding) to the positive uncoated portion **111a** and the negative electrode uncoated portion **112a**, respectively.

[0076] In one or more embodiments, the positive and negative electrode terminals **121** and **122** may be coupled by welding to the positive and negative electrode current collectors **140** and **150**, respectively. However, the present disclosure is not limited thereto, and the positive and negative electrode terminals **121** and **122** and the positive and negative electrode current collectors **140** and **150** may be integrally formed in one or more embodiments.

[0077] In addition, an insulation unit may be between the electrode assembly **110** and the cap plate **131**. The insulation unit may include first and second lower insulation units **160** and **170**, and each of the first and second lower insulation units **160** and **170** may also have a portion located between the electrode assembly **110** and the case **120**.

[0078] In addition, according to one or more embodiments of the present disclosure, one end of a separation member may face one side of the electrode assembly **110** and may be installed between the insulation unit and the positive or negative electrode terminals **121** and **122**.

[0079] In one or more embodiments, the separation member may include first and second separation members **180** and **190**.

[0080] In one or more embodiments, first ends of the first and second separation members **180** and **190** facing one side of the electrode assembly **110** may be between a respective one of the first and second lower insulation units **160** and **170** and a respective one of the positive and negative electrode terminals **121** and **122**.

[0081] Accordingly, the positive and negative electrode terminals **121** and **122**, which may be coupled by welding to the positive and negative electrode current collectors **140** and **150**, may be

coupled to first ends of the first and second lower insulation units **160** and **170**, respectively, and the first and second separation members **180** and **190**, respectively.

[0082] FIG. **2** is a perspective view illustrating a battery module according to one or more embodiments of the present disclosure.

[0083] Referring to FIG. **2**, the battery module **200** according to one or more embodiments of the present disclosure includes electrode units **111** and **122**, a plurality of battery cells **100** arranged in one direction, a plurality of connection tabs **220** each connecting a battery cell **100a** to an adjacent battery cell **100b**, and a protection circuit module **230** having one end connected to the connection tabs **220**.

[0084] The protection circuit module **230** may include a battery management system (BMS). Further, the connection tab **220** may include a body portion in contact with the electrode units **121** and **122** between the adjacent battery cells **100a** and **100b** and an extension portion extending from the body portion and connected to the protection circuit module **230**. The connection tab **220** may be, for example, a bus bar.

[0085] Each battery cell **100** may include a battery case, an electrode assembly received (or accommodated) in the battery case, and an electrolyte. The electrode assembly and the electrolyte react electrochemically to store and release (e.g., generate) energy. Terminal parts **121** and **122** electrically connected to the connection tab **220** and a vent **134** configured to discharge gas generated inside the battery case may be provided on one side of (e.g., an upper side of) the battery cell **100**. The terminal parts **121** and **122** of the battery cell **100** may be a positive electrode terminal **121** and a negative electrode terminal **122** having different polarities from each other, and the terminal parts **121** and **122** of the adjacent battery cells **100a** and **100b** may be electrically connected to each other in series or parallel by the connection tab **220**, to be described in more detail below. Although a serial connection has been described as an embodiment, the connection structure is not limited thereto, and various connection structures may be employed as desired or necessary. In addition, the number and arrangement of battery cells is not limited to the structure shown in FIG. **2** and may be changed as desired or necessary.

[0086] The plurality of battery cells **100** may be arranged in (e.g., may be stacked in) one direction so that the wide surfaces of the battery cells **100** face each other, and the plurality of battery cells **100** may be fixed by the housings **261**, **262**, **263**, and **264**. The housings **261**, **262**, **263**, and **264** may include a pair of end plates **261** and **262** facing the wide surfaces of the endmost battery cells **100** and a side plate **263** and a bottom plate **264** connecting the pair of end plates **261** and **262** to each other. The side plate **263** may support side surfaces of the battery cells **100**, and the bottom plate **264** may support bottom surfaces of the battery cells **100**. In addition, the pair of end plates **261** and **262**, the side plate **263**, and the bottom plate **264** may be connected by fasteners (e.g., bolts) **265** and/or any other suitable fastening members and methods known to those of ordinary skill in the art.

[0087] The protection circuit module **230** may have electronic components and protection circuits mounted thereon and may be electrically connected to the connection tabs **220**, to be described in more detail later. The protection circuit module **230** includes a first protection circuit module **230a** and a second protection circuit module **230b** extending along the direction in which the plurality of battery cells **100** are arranged (e.g., the Y-axis direction). The first protection circuit module **230a** and the second protection circuit module **230b** may be spaced from each other at a suitable interval (e.g., a predetermined interval in the X-axis direction) and arranged parallel (or substantially parallel) to each other and electrically connected to adjacent connection tabs **220**. In one or more embodiments, the first protection circuit module **230a** extends along one side of the upper portion of the plurality of battery cells **100** in the direction in which the plurality of battery cells **100** are arranged (e.g., the Y-axis direction), and the second protection circuit module **230b** extends along the other side of the upper portion of the plurality of battery cells **100** in the direction in which the plurality of battery cells **100** are arranged (e.g., the Y-axis direction). The second protection circuit

module **230b** may be spaced from the first protection circuit module **230a** at a suitable interval (e.g., a predetermined interval in the X-axis direction) with the vents **134** of the battery cells **100** between the first and second protection circuit modules **230a**, **230b**. The vents **134** may parallel (or substantially parallel) to the first and second protection circuit module **230a**, **230b**. As such, the two protection circuit modules are spaced from each other side-by-side along the direction in which the plurality of battery cells **100** are arranged, thereby reducing or minimizing the area of the printed circuit board (PCB) constituting the protection circuit module. By separately configuring the protection circuit module into two protection circuit modules, unnecessary PCM area can be reduced or minimized. In addition, the first protection circuit module **230a** and the second protection circuit module **230b** may be connected to each other by a conductive connection member **250**. One side of the conductive connection member **250** is connected to the first protection circuit module **230a**, and the other side thereof is connected to the second protection circuit module **230b** so that the two protection circuit modules **230a** and **230b** can be electrically connected with each other.

[0088] The connection may be performed by any one of soldering, resistance welding, laser welding, projection welding and/or any other suitable connection methods known to those of ordinary skill in the art.

[0089] In addition, the connection member **250** may be, for example, an electric wire. In addition, the connection member **250** may be made of a material having elasticity or flexibility. Due to the connecting member **250**, it may be possible to check and manage whether the voltage, temperature, and/or current of the plurality of battery cells **100** are normal. In one or more embodiments, the information received by the first protection circuit module from connection tabs adjacent to the first protection circuit module, such as voltage, current, and/or temperature, and the information received from connection tabs adjacent to the second protection circuit module, such as voltage, current, and/or temperature, may be integrated and managed by the protection circuit module through the connection member **50**.

[0090] In addition, in response to the battery cell **100** swelling, shocks may be absorbed by the elasticity or flexibility of the connection member **250**, thereby preventing (or at least mitigating) the first and second protection circuit modules **230a** and **230b** from being damaged.

[0091] In addition, the shape and structure of the connection member **250** is not limited to the shape and structure shown in FIG. 2.

[0092] As described above, because the protection circuit module **230** is provided as the first and second protection circuit modules **230a** and **230b**, the area of the PCB constituting the protection circuit module can be reduced or minimized, and the space inside the battery module can be secured, which improves work efficiency by facilitating a fastening work for connecting the connection tabs **220** and the protection circuit module **30** and repair work in response to an abnormality being detected in the battery module.

[0093] FIGS. 3A and 3B show a battery pack according to one or more embodiments of the present disclosure.

[0094] The battery pack may include a plurality of battery modules **200** and a housing **310** for accommodating the plurality of battery modules **200**. In one or more embodiments, the housing **310** may include first and second housings **311** and **312** coupled in opposite directions through the plurality of battery modules **200**. The plurality of battery modules **200** may be electrically connected to each other by using a bus bar **251**, and the plurality of battery modules **200** may be electrically connected to each other in a series/parallel or series-parallel mixed method, thereby obtaining desired (e.g., required) electrical output.

[0095] FIGS. 4A and 4B show a vehicle body and vehicle body parts having a battery pack according to one or more embodiments of the present disclosure.

[0096] In FIG. 4A, a battery pack **300** may include a battery pack cover **311**, which is a part of a vehicle underbody **410**, and a pack frame **312** under the vehicle underbody **410**. The pack frame

312 and the battery pack cover **311** may be integral with a vehicle floor **420**.

[0097] The vehicle underbody **410** separates the inside and outside of a vehicle, and the pack frame **312** may be outside the vehicle.

[0098] FIG. **4B** is a schematic side view of a vehicle **500** according to one or more embodiments of the present disclosure.

[0099] A vehicle **500** may be formed by combining additional parts, such as a hood **510** in front of the vehicle and fenders **520** respectively located in the front and rear of the vehicle to a vehicle body **400**.

[0100] The vehicle **500** may further include a vehicle floor **420**, which is one of the vehicle body parts **400** including the battery pack **300**, which includes the pack frame **312** and the battery pack cover **311**.

[0101] FIGS. **5A** to **5D** illustrate a method for charging a secondary battery according to an embodiment of the present disclosure. A secondary battery may be charged and discharged in, for example, the following manner:

CCCV (Constant Current Constant Voltage) Charging

[0102] CCCV charging is a charging method in which constant current (CC) charging is performed until the voltage reaches a suitable or reference voltage (e.g., a predetermined voltage), and then constant voltage (CV) charging is performed until the amount of current flowing decreases until the end current value is reached.

[0103] During CC charging, as shown in FIG. **5A**, a switch of a constant current power supply is turned on and a switch of a constant voltage power supply is turned off so that a constant current I flows through the secondary battery. During CC charging, because the current I is constant, a voltage V_R applied to an internal resistor R is also constant according to Ohm's law ($V_R = R \times I$). Additionally, during the CC charging, a voltage V_C applied to a secondary battery capacitor C increases with time. Therefore, a secondary battery voltage V_B may rise over time.

[0104] In response to the secondary battery voltage V_B reaching a suitable or reference voltage (e.g., a predetermined voltage), for example, about (approximately) 4.3 V, CC charging is switched to CV charging. During CV charging, as shown in FIG. **5B**, the switch of the constant voltage power supply is turned on and the switch of the constant current power supply is turned off so that the secondary battery voltage V_B is constant (or substantially constant). Additionally, during the CV charging, the voltage V_C applied to the secondary battery capacitor C increases with time. Because $V_B = V_R + V_C$ is satisfied, the voltage V_R applied to the internal resistor R decreases with time. As the voltage V_R applied to the internal resistor R decreases, the current I flowing through the secondary battery may also decrease according to Ohm's law ($V_R = R \times I$).

[0105] In response to the current I flowing through the secondary battery reaching a suitable or reference current (e.g., a predetermined current), for example, about (approximately) 0.01 C, charging is terminated. In response to the CCCV charging finishing, all switches are turned off and the current I becomes 0 (or substantially 0), as shown in FIG. **5C**. At this time, the voltage V_R applied to the internal resistor R becomes 0 V (or substantially 0 V). Accordingly, even when voltage drop is prevented across the internal resistor R , the secondary battery voltage V_B does not decrease (or at least substantially does not decrease).

[0106] FIG. **5D** shows a graph of a secondary battery voltage V_B and charging current during CCCV charging and after CCCV charging is terminated. Even after CCCV charging is terminated, the secondary battery voltage V_B does not decrease (or at least substantially does not decrease).

[0107] FIG. **6** is a partial exploded perspective view of a vent unit **134** and an insulating member **135** of a secondary battery **100** according to an embodiment of the present disclosure. FIG. **7A** is a partial sectional view of the vent unit **134** and the insulating member **135** of the secondary battery **100** according to the embodiment of the present disclosure. FIG. **7B** is a partial sectional view showing the vent unit **134** of FIG. **7A** in a ruptured state. FIG. **8** is a partial exploded perspective view of a vent unit **134** and an insulating member **136** of a secondary battery **100** according to

another embodiment of the present disclosure. FIG. 9 is a partial sectional view of the vent unit **134** and the insulating member **136** of the secondary battery **100** according to the other embodiment of the present disclosure. Herein, “battery cell” and “secondary battery” may be used interchangeably. [0108] Referring to FIGS. 1A, 1B, 2, and 6, the secondary battery **100** may include a case **120** having a space defined therein, an electrode assembly **110** accommodated in the space of the case **120**, a cap assembly **130**, and an insulating member **135** at the cap assembly **130**. In one or more embodiments, the cap assembly **130** may include a cap plate **131** configured to seal the space of the case **120**, the cap plate **131** including a vent unit **134** in one surface, and terminals **121** and **122** coupled to one side of the cap plate **131** and electrically connected to the electrode assembly **110**. [0109] In one or more embodiments, the insulating member **135** may be located so as to correspond to the vent unit **134** and may be configured to cover at least a part (or portion) of the vent unit **134**. In one or more embodiments, the insulating member **135** may be at an upper side of the vent unit **134** and cover the edge of the vent unit **134**, thereby preventing (or at least mitigating) damage around the vent unit **134** during heat propagation.

[0110] In one or more embodiments, the cap plate **131** may include a vent hole **1311** through which the vent unit **134** and the insulating member **135** are coupled to each other, and a seating recess **1312** to which the insulating member **135** is coupled. The seating recess **1312** may be in an upper surface of the cap plate **131** and extend around the vent hole **1311**.

[0111] In one or more embodiments, the vent unit **134** may be coupled or connected to the cap plate **131** and seal the vent hole **1311** in one side of the cap plate **131**. The shape of the vent hole **1311** and the shape of the vent unit **134** may correspond to each other (e.g., the vent hole **1311** and the vent unit **134** may have the same shape). As used herein, “corresponding” means that two parts are similar or identical to each other. In one or more embodiments, both the vent hole **1311** and the vent unit **134** may be oval. In one or more embodiments, the vent unit **134** may include a connection portion **134b** overlapping the cap plate **131** and fixed to the cap plate **131** and a rupture portion **134a** inside the connection portion **134b** that includes a notch. In one or more embodiments, the connection portion **134b** may be the edge portion of the vent unit **134**. The connection portion **134b** may be coupled to a lower end of the vent hole **1311**.

[0112] In one or more embodiments, the insulating member **135** may have an opening **1353** in the center region (or substantially the center region) to expose the vent unit **134**. In response to the vent unit **134** being ruptured, at least a part of the vent unit **134** may protrude outwardly through the opening **1353**. In one or more embodiments, the insulating member **135** may include an insulating plate **1351** coupled to the upper surface of the cap plate **131** and a protrusion **1352** protruding (e.g., downward) from the insulating plate **1351** toward the vent unit **134**. In one or more embodiments, the insulating plate **1351** may be coupled to the seating recess **1312** of the cap plate **131**. The protrusion **1352** may be coupled to the vent hole **1311** of the cap plate **131**. The protrusion **1352** may correspond to the shape of the vent hole **1311**. In one or more embodiments, the opening **1353** may extend through the insulating plate **1351** and the protrusion **1352**. In one or more embodiments, the protrusion **1352** may have a hollow, columnar structure corresponding to the shape of the vent hole **1311**. The insulating plate **1351** may have a flat structure with a hollow center region corresponding to the shape of the vent hole **1311**. However, the present disclosure is not limited thereto, and the insulating member **135** may have a ring-shaped top structure with a hollow interior. In one or more embodiments, the insulating plate **1351** may not have a plate shape. [0113] In one or more embodiments, referring to FIG. 7A, a vent unit **134** may be at (e.g., on) a lower surface of the cap plate **131**, and the insulating member **135** may be coupled to the upper surface of the cap plate **131** at a position corresponding to the position of the vent unit **134**. The insulating member **135** may be coupled along the edge of the vent hole **1311** through the protrusion **1352**.

[0114] In one or more embodiments, the insulating member **135** may be coupled to the seating recess **1312** of the cap plate **131** with an interference fit. In one or more embodiments, the

insulating member **135** may be coupled to the upper surface of the cap plate **131** by thermal fusion. The insulating member **135** and the cap plate **131** may be integrally formed by double injection molding. In one or more embodiments, the upper surface of the insulating member **135** and the upper surface of the cap plate **131** may be on the same plane (or substantially the same plane) (e.g., the upper surface of the insulating member **135** and the upper surface of the cap plate **131** may be co-planar or substantially co-planar). In one or more embodiments, the insulating member **135** may include at least one of polyphenylene sulfide (PPS), polyetheretherketone (PEEK), and/or mica. In one or more embodiments, the insulating member **135** may further include a high heat resistant resin such as polyphthalamide (PPA), thermoplastic polyimide (TPI), polysulfone (PSU), polyethersulfone (PES), polyetherimide, and/or liquid crystal polymer (LCP). In one or more embodiments, mica may be provided in a sheet form. In one or more embodiments, the mica sheet may include a rigid mica sheet (e.g., phlogopite or muscovite), a composite mica sheet (e.g., glass fiber reinforced mica sheet), flexible mica (e.g., rolled mica), or a molded mica sheet.

[0115] Referring to FIG. 7B, in some embodiments, in response to the internal pressure of the case **120** being greater than a preset threshold pressure and the rupture portion **134a** being opened (i.e., an event occurs), the rupture portion **134a** may rupture toward the edge of the opening **1353** of the insulating member **135**. Consequently, the insulating member **135** may be fixed by the deformed vent unit **134**.

[0116] In one or more embodiments, damage around the vent of the battery cell where the event occurred may be prevented (or at least mitigated), and heat propagation may be remedied by reducing thermal effects through the cap plate around the vent of the battery cell.

[0117] Referring to FIGS. 8 and 9, the insulating member **136** may be configured to cover the entirety (or substantially the entirety) of the vent unit **134**. The insulating member **136** may have a flat (planar) shape so as to be coupled to the seating recess **1312**, rather than having the structure of the protrusion **1352** formed separately. Consequently, a gap may be provided between the insulating member **136** and the vent unit **134**. In some embodiments, the upper surface of the insulating member **136** and the upper surface of the cap plate **131** may be on the same or substantially the same plane (i.e., co-planar or substantially co-planar). Configurations other than the structure of the insulating member **136** are identical to those described above, and therefore a description thereof will be omitted.

[0118] As is apparent from the above description, embodiments provide a secondary battery capable of preventing (or at least mitigating) damage around a cell vent and a secondary battery module including the same.

[0119] Embodiments provide a secondary battery capable of preventing (or at least mitigating) heat propagation and thermal runaway and a secondary battery module including the same.

[0120] Although the present disclosure has been described by limited embodiments and drawings, the present disclosure is not limited thereto. However, various modifications and variations can be made by a person skilled in the art to which the present disclosure belongs within the scope of equivalency of the technical idea of the present disclosure and the scope of the present invention as set forth in the following claims.

Claims

1. A secondary battery comprising: a case comprising a space; an electrode assembly accommodated in the space of the case; a cap assembly comprising a cap plate configured to seal the space of the case, the cap plate comprising a vent unit in one surface of the cap plate, and a terminal coupled to one side of the cap plate, the terminal being electrically connected to the electrode assembly; and an insulating member coupled to the surface of the cap plate, the insulating member being configured to cover at least a part of the vent unit.
2. The secondary battery as claimed in claim 1, wherein: the vent unit is coupled to a lower surface

of the cap plate, and the insulating member is coupled to an upper surface of the cap plate corresponding to a position of the vent unit.

3. The secondary battery as claimed in claim 1, wherein: the vent unit is coupled to the cap plate, the vent unit sealing a vent hole in one side of the cap plate, and the vent unit comprises a connection portion overlapping the cap plate, the connection portion being fixed to the cap plate, and a rupture portion inside the connection portion, the rupture portion comprising a notch.

4. The secondary battery as claimed in claim 3, wherein the insulating member comprises an opening in a center region of the insulating member, the opening being configured to expose the vent unit.

5. The secondary battery as claimed in claim 4, wherein the insulating member comprises an insulating plate coupled to an upper surface of the cap plate and a protrusion protruding from the insulating plate toward the vent unit.

6. The secondary battery as claimed in claim 5, wherein the protrusion corresponds to a shape of the vent hole.

7. The secondary battery as claimed in claim 5, wherein the protrusion comprises a hollow, columnar structure corresponding to a shape of the vent hole.

8. The secondary battery as claimed in claim 5, wherein the opening extends through the insulating plate and the protrusion.

9. The secondary battery as claimed in claim 5, wherein the insulating member is coupled along an edge of the vent hole through the protrusion.

10. The secondary battery as claimed in claim 5, wherein the cap plate further comprises a seating recess in the upper surface of the cap plate around the vent hole.

11. The secondary battery as claimed in claim 10, wherein the insulating plate is coupled to the seating recess.

12. The secondary battery as claimed in claim 11, wherein the insulating member is coupled to the seating recess of the cap plate by an interference fit.

13. The secondary battery as claimed in claim 1, wherein the insulating member comprises a ring-shaped top structure with a hollow interior.

14. The secondary battery as claimed in claim 1, wherein the insulating member is thermally fused to an upper surface of the cap plate.

15. The secondary battery as claimed in claim 1, wherein the insulating member and the cap plate are integral by double injection molding.

16. The secondary battery as claimed in claim 1, wherein an upper surface of the insulating member and an upper surface of the cap plate are co-planar.

17. The secondary battery as claimed in claim 1, wherein the insulating member comprises at least one of polyphenylene sulfide (PPS), polyetheretherketone (PEEK), or mica.

18. The secondary battery as claimed in claim 4, wherein the rupture portion is configured to rupture along an edge of the opening of the insulating member in response to an internal pressure of the case.

19. The secondary battery as claimed in claim 1, wherein the insulating member is configured to cover substantially an entirety of the vent unit.

20. A secondary battery module comprising: a plurality of secondary batteries arranged in one direction; a housing comprising sidewalls forming a receiving portion configured to accommodate the plurality of secondary batteries; and a busbar configured to electrically connect neighboring ones of the plurality of secondary batteries to each other, wherein each of the plurality of secondary batteries comprises: a case comprising a space; an electrode assembly accommodated in the space; a cap assembly comprising a cap plate configured to seal the space of the case, the cap plate comprising a vent unit in one surface of the cap plate, and a terminal coupled to one side of the cap plate, the terminal being electrically connected to the electrode assembly; and an insulating

member coupled to the surface of the cap plate, the insulating member being configured to cover at least a part of the vent unit.
